TASHVI SINGLA

tashvi225@gmail.com | New York, NY | Houston, TX | (346) 387 4074

EDUCATION

Columbia University, B.S. in Mechanical Engineering | Minor: Aerospace Engineering | GPA: 3.8/4.0 Expected 2029
Coursework (In Progress): Thermodynamics, Solid Mechanics, Space Flight Mechanics, ODEs, Fluid Mechanics

EXPERIENCE

Zeta Energy Corporation Jun 2026 - Sep 2026
Mechanical Engineering Intern *Houston, TX*
- Engineered a custom lithium-sulfur battery pack achieving a 100% improvement in Wh/kg at equivalent weight; validated through qualification and flight testing
- Designed structural mounting, wiring, and connector integration for a custom battery pack electronics assembly, engineering the layout around thermal and safety constraints to protect structural and system reliability under real-world operating loads

Columbia Formula SAE Sep 2025 - Present
Manufacturing Team | Dynamics Team *New York, NY*
- Fabricating the nose cone and ergo jig of the FSAE car using carbon fiber molding and CNC machining
- Redesigned suspension rod in SolidWorks/CAD to reduce unsprung mass while increasing torsional stiffness; validated structural analysis using first principles and FEA

Columbia Space Initiative Sep 2025 - Present
Rockets Propulsion-Fluids Team | Roboglove Team *New York, NY*
- Developing gearbox for electronic regulator feed system valve for a hybrid rocket engine using liquid oxygen
- Designed and fabricated structural linkage of a tendon-driven robotic hand exoskeleton for space suit integration; iterated tendon routing across prototypes, increasing grip force 25% and cutting dexterity task time 15%

Columbia Robotics Sep 2026 - Present
Drones Team *New York, NY*
- Designed frame, arms, propellers, and snake-style payload mechanism for a quadcopter in Onshape; 3D printed a prototype arm and propellers to validate fit and stiffness ahead of full build

Columbia Engineers Without Borders Sep 2025 - Present
Funding & Grants Lead - Mpakadan, Ghana *New York, NY*
- Conducted on-site assessment evaluating electrical grid, power stability, transformer, and system performance and reliability against design specifications
- Authored wiring schematics, load calculations, diagrams, and full technical documentation of electrical design and installation processes, to support reliable power delivery; secured $5,000 in fundraisers & grants

FIRST FRC Robotics Team #8576 Jan 2022 - May 2025
Vice President (Gr. 11-12) | Fundraising & Finance Officer (Gr. 11) *Fulshear, TX*
- Managed 6-week build cycle from CAD to competition-ready hardware for a 126-member team as primary design lead, coordinating fabrication scheduling, part quality checks, and assembly troubleshooting; won Texas State Championship

PERSONAL PROJECTS

Model Rocket - Aerodynamic Design & Stability Testing
Constructed and laser-cut a balsa wood airframe with a 3D-printed nose cone and spring-loaded payload release mechanism; performed hand calculations and structural analysis on fin geometry and static margin. Predicted altitude/trajectory via OpenRocket; validated CG/CP placement against flight testing.

Drone Platform - Airframe, Propulsion, & Power System Design
Designed propulsion layout, airframe structure, and power system integration to enable extended-endurance autonomous UAV operation, supporting onboard sensor and compute payload requirements.

Multi-Stage Gear Train - Design, Fabrication & Efficiency Testing
Engineered a multi-stage gear train in SolidWorks and fabricated via 3D printing; analytically predicted gear ratio and torque output; measured mechanical efficiency experimentally using a calibrated moment arm across multiple loading conditions.

SKILLS

CAD / Manufacturing: SolidWorks, Onshape, FEA Simulation, DFM, Structural Analysis, 3D Printing, Laser Cutting, CNC
Programming & Tools: Python, Java, SQL, R Programming, Excel Programming, Mission Planner, OpenRocket, Arduino